

3D Transformation in Computer Graphics-

In Computer graphics,

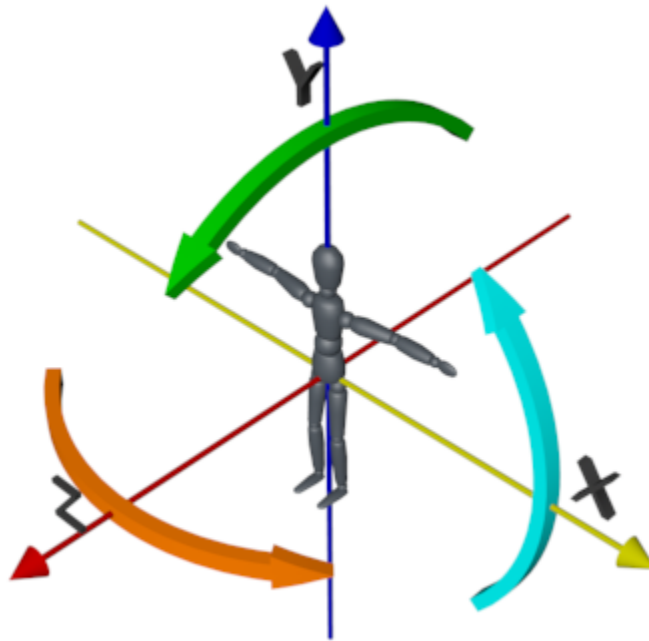
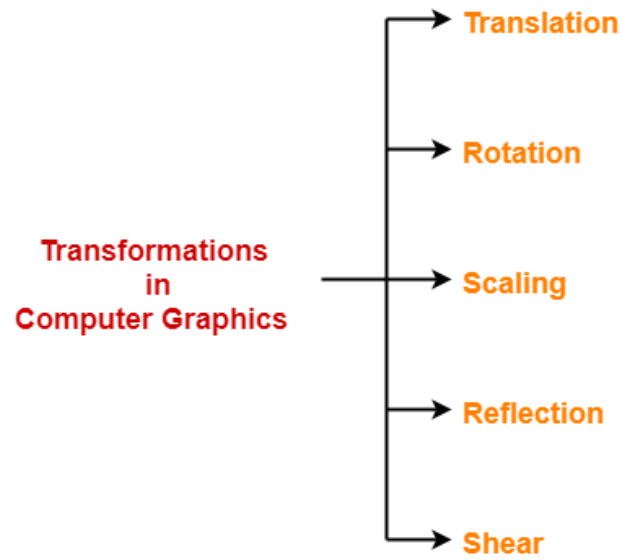
Transformation is a process of modifying and re-positioning the existing graphics.

- There are two types of transformation in computer graphics.
 - 1) 2D transformation
 - 2) 3D transformation
- 3D Transformations take place in a three dimensional plane.
- 3D Transformations are important and a bit more complex than 2D Transformations.
- Transformations are helpful in changing the position, size, orientation, shape etc of the object.

<https://technology.cpm.org/general/3dgraph/>

Transformation Techniques-

In computer graphics, various transformation techniques are-

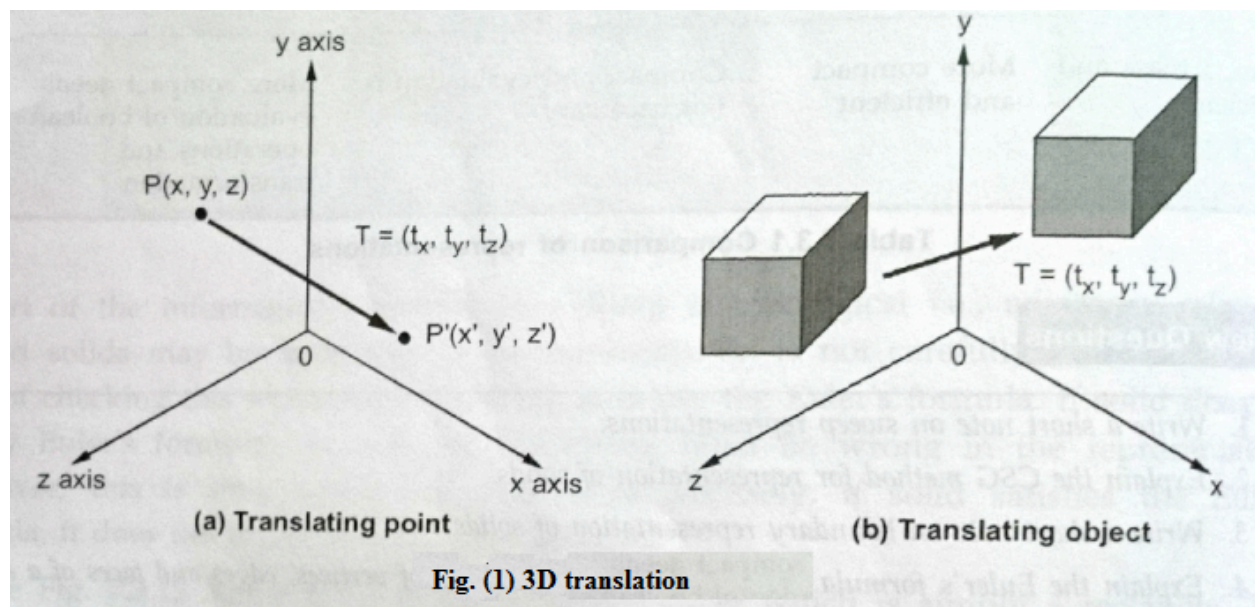
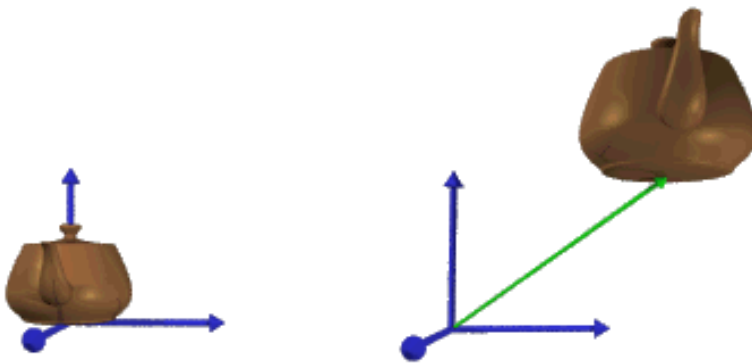


3D Translation in Computer Graphics-

In Computer graphics,

3D Translation is a process of moving an object from one position to another in a three dimensional plane.

Consider a point object O has to be moved from one position to another in a 3D plane.

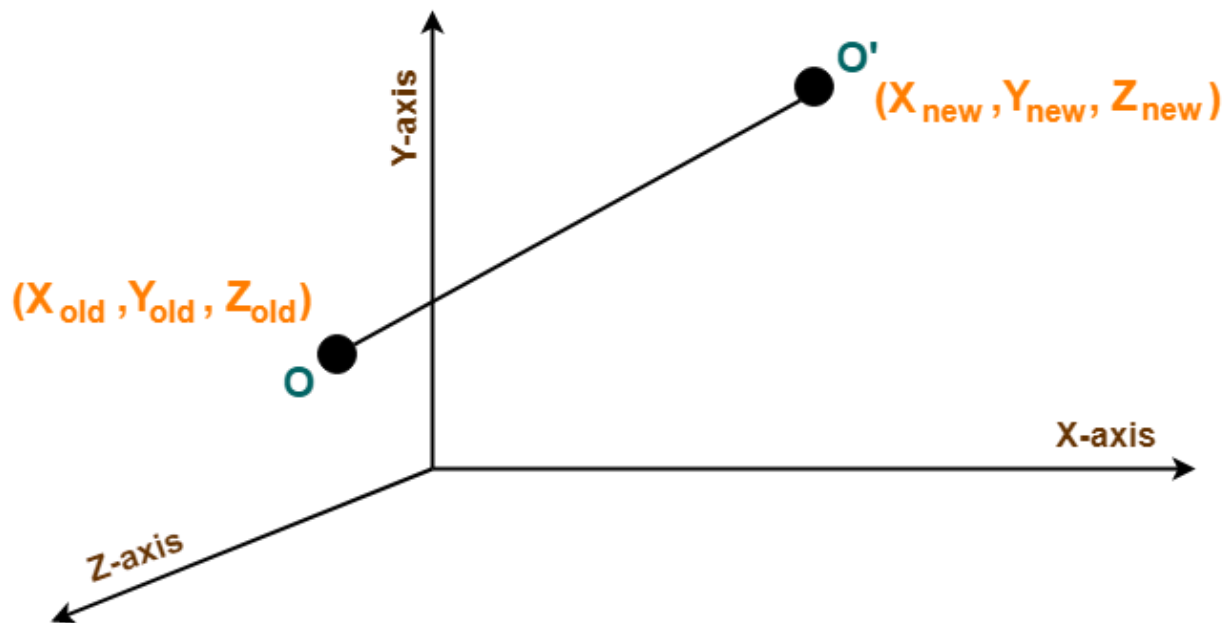


Let-

- Initial coordinates of the object $O = (X_{old}, Y_{old}, Z_{old})$
- New coordinates of the object O after translation = $(X_{new}, Y_{new}, Z_{old})$
- Translation vector or Shift vector = (T_x, T_y, T_z)

Given a Translation vector (T_x, T_y, T_z) -

- T_x defines the distance the X_{old} coordinate has to be moved.
- T_y defines the distance the Y_{old} coordinate has to be moved.
- T_z defines the distance the Z_{old} coordinate has to be moved.



3D Translation in Computer Graphics

This translation is achieved by adding the translation coordinates to the old coordinates of the object as-

- $X_{new} = X_{old} + T_x$ (This denotes translation towards X axis)
- $Y_{new} = Y_{old} + T_y$ (This denotes translation towards Y axis)
- $Z_{new} = Z_{old} + T_z$ (This denotes translation towards Z axis)

Problem-

Given a 3D object with coordinate points A(0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0). Apply the translation with the distance 1 towards X axis, 1 towards Y axis and 2 towards Z axis and obtain the new coordinates of the object.

Solution-

Given-

- Old coordinates of the object = A (0, 3, 1), B(3, 3, 2), C(3, 0, 0), D(0, 0, 0)
- Translation vector = $(T_x, T_y, T_z) = (1, 1, 2)$

For Coordinates A(0, 3, 1)

Let the new coordinates of A = $(X_{new}, Y_{new}, Z_{new})$.

Applying the translation equations, we have-

- $X_{new} = X_{old} + T_x = 0 + 1 = 1$
- $Y_{new} = Y_{old} + T_y = 3 + 1 = 4$
- $Z_{new} = Z_{old} + T_z = 1 + 2 = 3$

Thus, New coordinates of A = (1, 4, 3).

For Coordinates B(3, 3, 2)

Let the new coordinates of B = $(X_{new}, Y_{new}, Z_{new})$.

Applying the translation equations, we have-

- $X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$
- $Y_{\text{new}} = Y_{\text{old}} + T_y = 3 + 1 = 4$
- $Z_{\text{new}} = Z_{\text{old}} + T_z = 2 + 2 = 4$

Thus, New coordinates of B = (4, 4, 4).

For Coordinates C(3, 0, 0)

Let the new coordinates of C = (X_{new} , Y_{new} , Z_{new}).

Applying the translation equations, we have-

- $X_{\text{new}} = X_{\text{old}} + T_x = 3 + 1 = 4$
- $Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$
- $Z_{\text{new}} = Z_{\text{old}} + T_z = 0 + 2 = 2$

Thus, New coordinates of C = (4, 1, 2).

For Coordinates D(0, 0, 0)

Let the new coordinates of D = (X_{new} , Y_{new} , Z_{new}).

Applying the translation equations, we have-

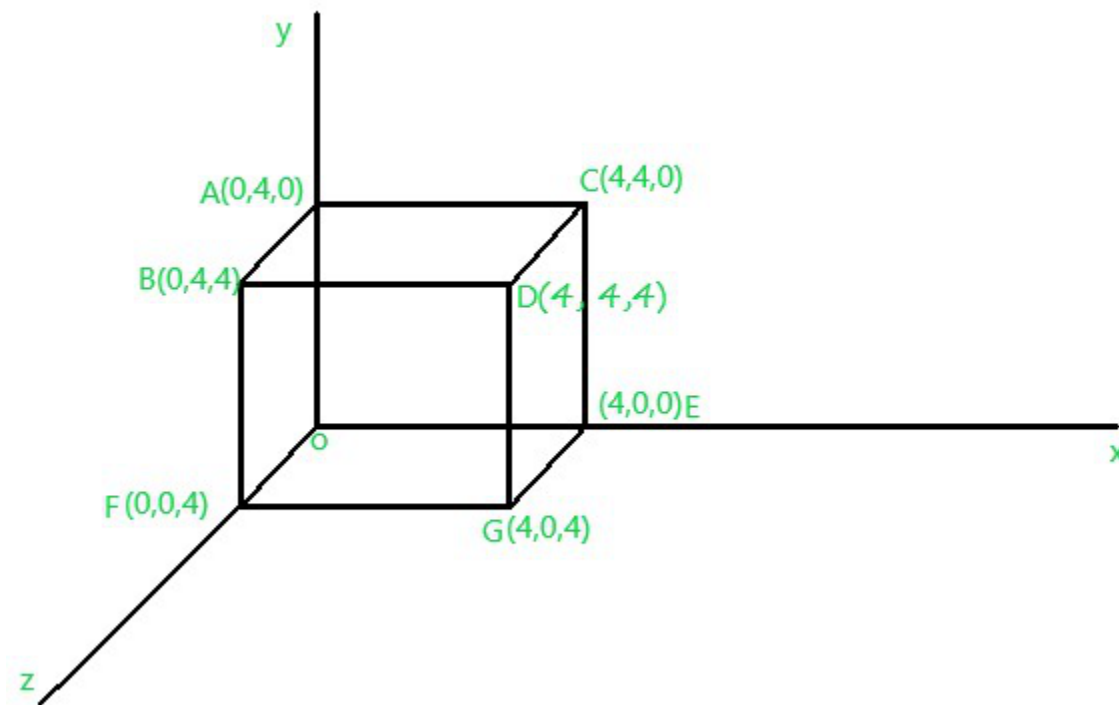
- $X_{\text{new}} = X_{\text{old}} + T_x = 0 + 1 = 1$
- $Y_{\text{new}} = Y_{\text{old}} + T_y = 0 + 1 = 1$
- $Z_{\text{new}} = Z_{\text{old}} + T_z = 0 + 2 = 2$

Thus, New coordinates of D = (1, 1, 2).

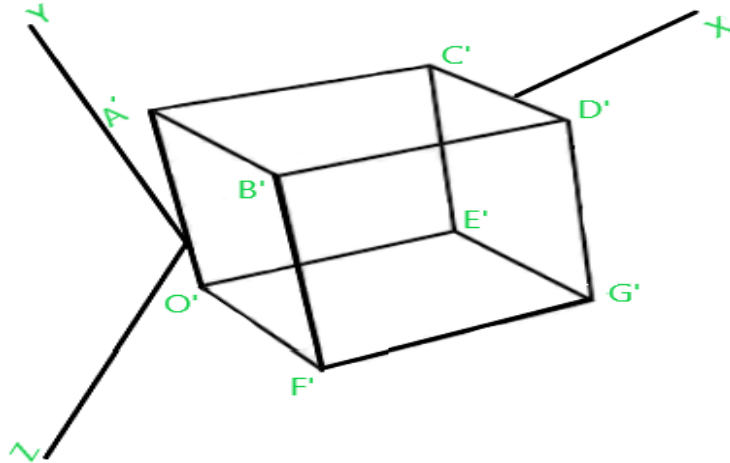
Thus, New coordinates of the object = A (1, 4, 3), B(4, 4, 4), C(4, 1, 2), D(1, 1, 2).

Problem 2 : Perform translation transformation on the following figure where the given translation distances are $T_x = 2$, $T_y = 4$, $T_z = 6$.

Solution : On applying Translation Transformation we get corresponding points –



After performing **translation transformation** over the **Fig.1**, it will look like as below –



3D Rotation in Computer Graphics-

In Computer graphics,

3D Rotation is a process of rotating an object with respect to an angle in a three dimensional plane.

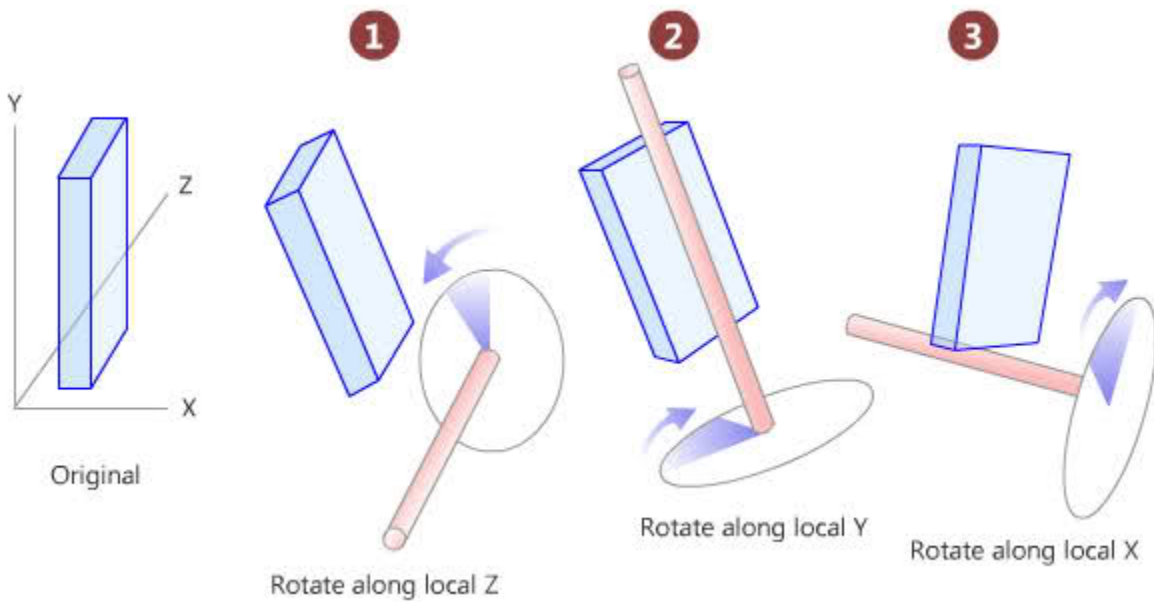
Consider a point object O has to be rotated from one angle to another in a 3D plane.

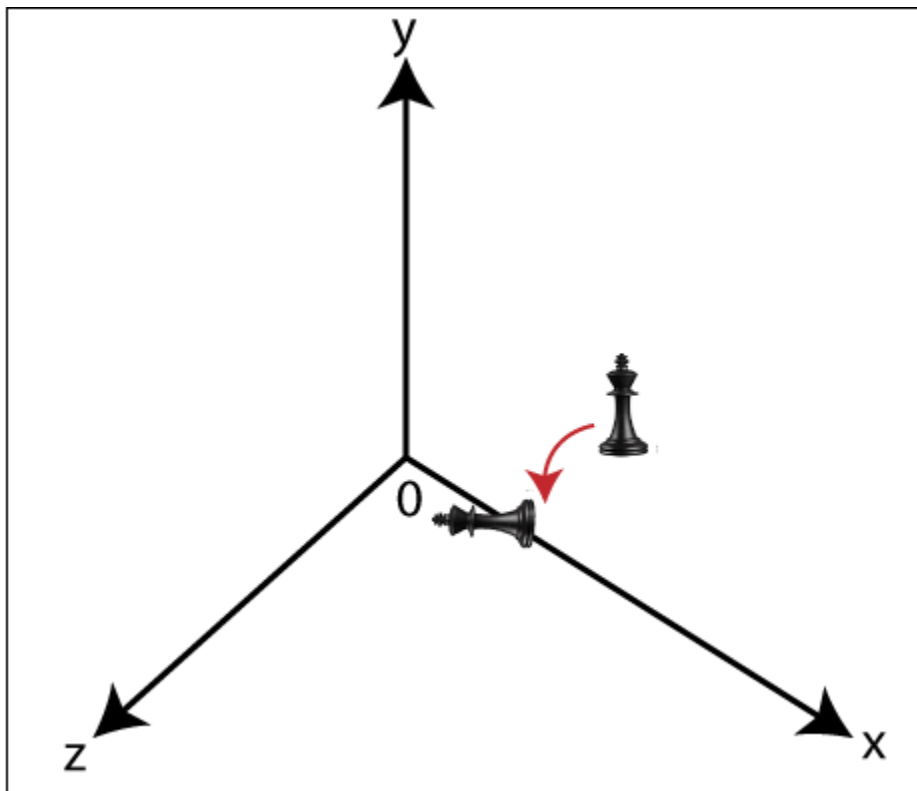
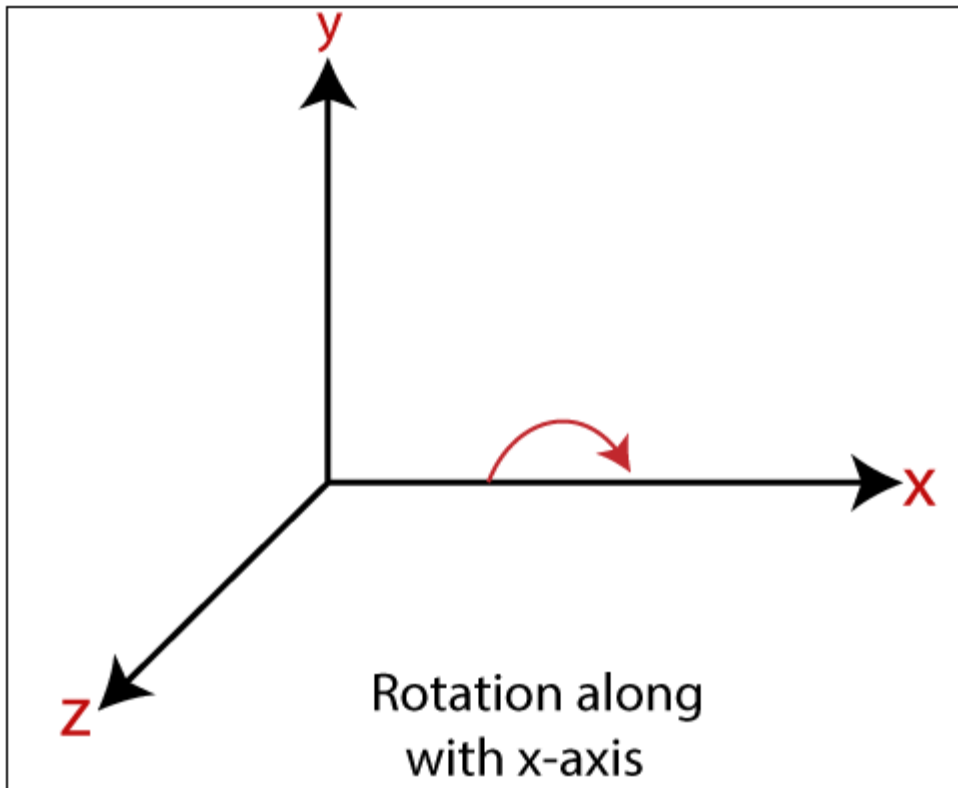
Let-

- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Initial angle of the object O with respect to origin = Φ
- Rotation angle = θ
- New coordinates of the object O after rotation = $(X_{new}, Y_{new}, Z_{new})$

In 3 dimensions, there are 3 possible types of rotation-

- X-axis Rotation
- Y-axis Rotation
- Z-axis Rotation

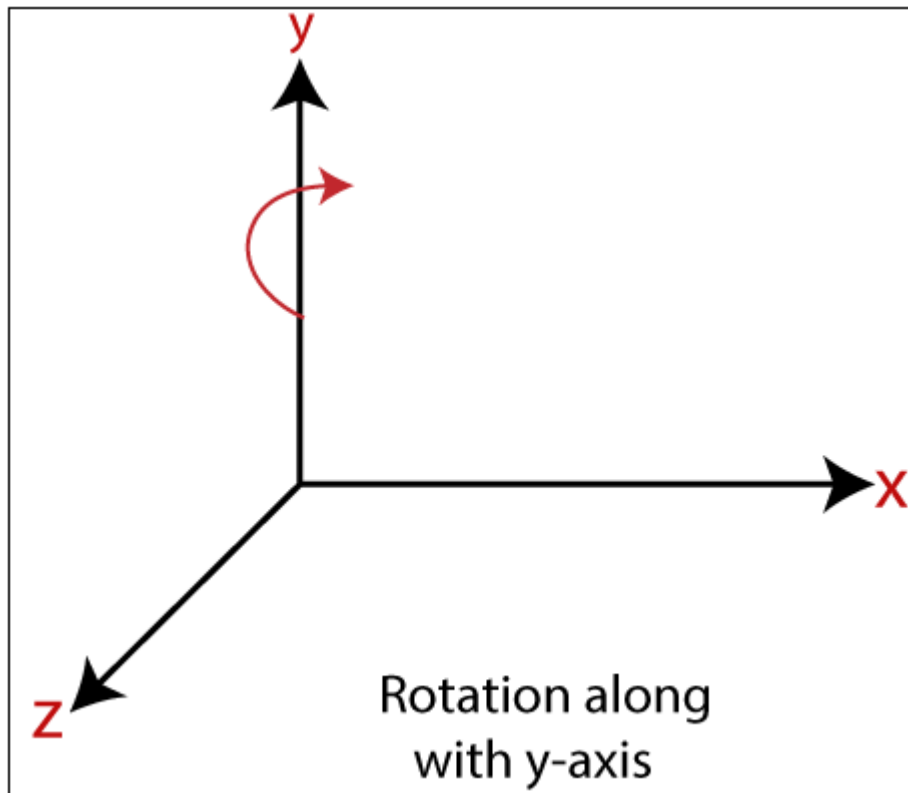


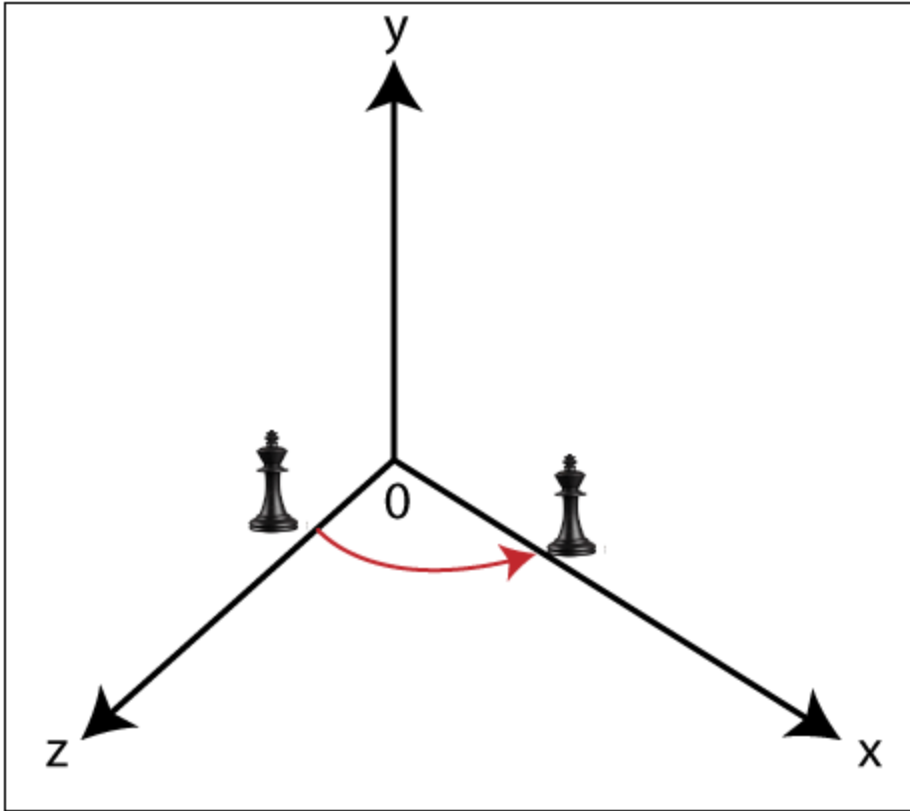


For X-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{\text{new}} = X_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}} \times \cos\theta - Z_{\text{old}} \times \sin\theta$
- $Z_{\text{new}} = Y_{\text{old}} \times \sin\theta + Z_{\text{old}} \times \cos\theta$

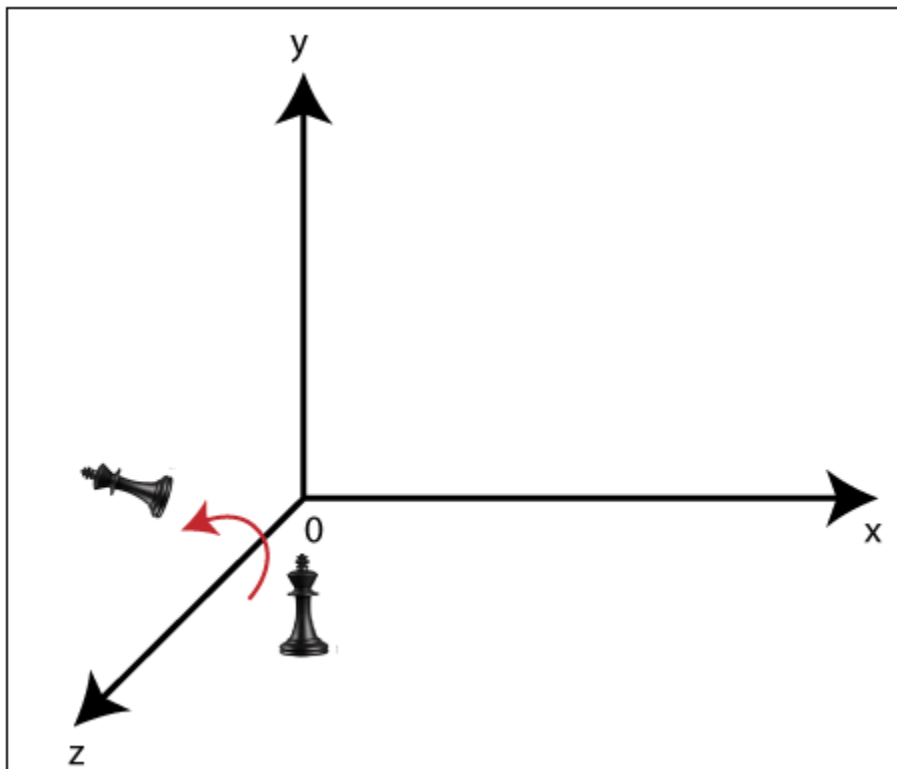
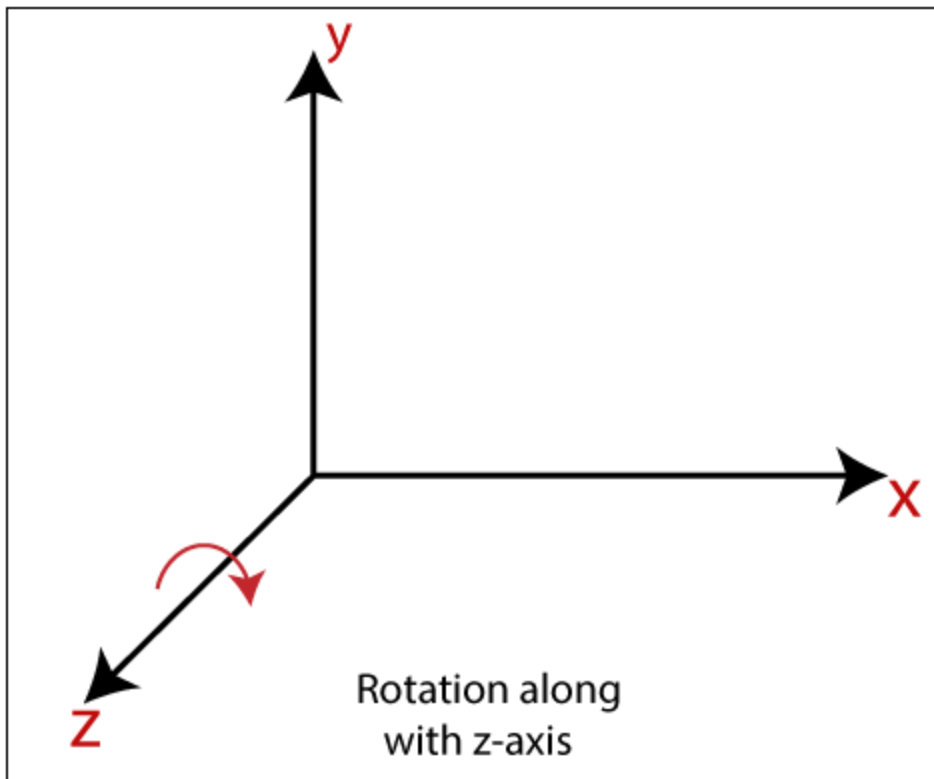




For Y-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{\text{new}} = Z_{\text{old}} \times \sin\theta + X_{\text{old}} \times \cos\theta$
- $Y_{\text{new}} = Y_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}} \times \cos\theta - X_{\text{old}} \times \sin\theta$



For Z-Axis Rotation-

This rotation is achieved by using the following rotation equations-

- $X_{\text{new}} = X_{\text{old}} \times \cos\theta - Y_{\text{old}} \times \sin\theta$
- $Y_{\text{new}} = X_{\text{old}} \times \sin\theta + Y_{\text{old}} \times \cos\theta$
- $Z_{\text{new}} = Z_{\text{old}}$

Problem-01:

Given a homogeneous point (1, 2, 3). Apply rotation 90 degree towards X, Y and Z axis and find out the new coordinate points.

Solution-

Given-

- Old coordinates = $(X_{\text{old}}, Y_{\text{old}}, Z_{\text{old}}) = (1, 2, 3)$
- Rotation angle = $\theta = 90^\circ$

For X-Axis Rotation-

Let the new coordinates after rotation = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the rotation equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 1$

- $Y_{\text{new}} = Y_{\text{old}} \times \cos\theta - Z_{\text{old}} \times \sin\theta = 2 \times \cos 90^\circ - 3 \times \sin 90^\circ = 2 \times 0 - 3 \times 1 = -3$
- $Z_{\text{new}} = Y_{\text{old}} \times \sin\theta + Z_{\text{old}} \times \cos\theta = 2 \times \sin 90^\circ + 3 \times \cos 90^\circ = 2 \times 1 + 3 \times 0 = 2$

Thus, New coordinates after rotation = (1, -3, 2).

For Y-Axis Rotation-

Let the new coordinates after rotation = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the rotation equations, we have-

- $X_{\text{new}} = Z_{\text{old}} \times \sin\theta + X_{\text{old}} \times \cos\theta = 3 \times \sin 90^\circ + 1 \times \cos 90^\circ = 3 \times 1 + 1 \times 0 = 3$
- $Y_{\text{new}} = Y_{\text{old}} = 2$
- $Z_{\text{new}} = Y_{\text{old}} \times \cos\theta - X_{\text{old}} \times \sin\theta = 2 \times \cos 90^\circ - 1 \times \sin 90^\circ = 2 \times 0 - 1 \times 1 = -1$

Thus, New coordinates after rotation = (3, 2, -1).

For Z-Axis Rotation-

Let the new coordinates after rotation = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the rotation equations, we have-

- $X_{\text{new}} = X_{\text{old}} \times \cos\theta - Y_{\text{old}} \times \sin\theta = 1 \times \cos 90^\circ - 2 \times \sin 90^\circ = 1 \times 0 - 2 \times 1 = -2$
- $Y_{\text{new}} = X_{\text{old}} \times \sin\theta + Y_{\text{old}} \times \cos\theta = 1 \times \sin 90^\circ + 2 \times \cos 90^\circ = 1 \times 1 + 2 \times 0 = 1$
- $Z_{\text{new}} = Z_{\text{old}} = 3$

Thus, New coordinates after rotation = (-2, 1, 3).

3D Scaling in Computer Graphics-

In computer graphics, scaling is a process of modifying or altering the size of objects.

- Scaling may be used to increase or reduce the size of object.
- Scaling subjects the coordinate points of the original object to change.
- Scaling factor determines whether the object size is to be increased or reduced.
- If scaling factor > 1 , then the object size is increased.
- If scaling factor < 1 , then the object size is reduced.

Consider a point object O has to be scaled in a 3D plane.

Let-

- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Scaling factor for X-axis = S_x
- Scaling factor for Y-axis = S_y
- Scaling factor for Z-axis = S_z
- New coordinates of the object O after scaling = $(X_{new}, Y_{new}, Z_{new})$

This scaling is achieved by using the following scaling equations-

- $X_{new} = X_{old} \times S_x$
- $Y_{new} = Y_{old} \times S_y$
- $Z_{new} = Z_{old} \times S_z$

Problem-01:

Given a 3D object with coordinate points A(0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0). Apply the scaling parameter 2 towards X axis, 3 towards Y axis and 3 towards Z axis and obtain the new coordinates of the object.

Solution-

Given-

- Old coordinates of the object = A (0, 3, 3), B(3, 3, 6), C(3, 0, 1), D(0, 0, 0)
- Scaling factor along X axis = 2
- Scaling factor along Y axis = 3
- Scaling factor along Z axis = 3

For Coordinates A(0, 3, 3)

Let the new coordinates of A after scaling = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the scaling equations, we have-

- $X_{\text{new}} = X_{\text{old}} \times S_x = 0 \times 2 = 0$
- $Y_{\text{new}} = Y_{\text{old}} \times S_y = 3 \times 3 = 9$
- $Z_{\text{new}} = Z_{\text{old}} \times S_z = 3 \times 3 = 9$

Thus, New coordinates of corner A after scaling = (0, 9, 9).

For Coordinates B(3, 3, 6)

Let the new coordinates of B after scaling = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the scaling equations, we have-

- $X_{\text{new}} = X_{\text{old}} \times S_x = 3 \times 2 = 6$
- $Y_{\text{new}} = Y_{\text{old}} \times S_y = 3 \times 3 = 9$
- $Z_{\text{new}} = Z_{\text{old}} \times S_z = 6 \times 3 = 18$

Thus, New coordinates of corner B after scaling = (6, 9, 18).

For Coordinates C(3, 0, 1)

Let the new coordinates of C after scaling = (X_{new} , Y_{new} , Z_{new}).

Applying the scaling equations, we have-

- $X_{\text{new}} = X_{\text{old}} \times S_x = 3 \times 2 = 6$
- $Y_{\text{new}} = Y_{\text{old}} \times S_y = 0 \times 3 = 0$
- $Z_{\text{new}} = Z_{\text{old}} \times S_z = 1 \times 3 = 3$

Thus, New coordinates of corner C after scaling = (6, 0, 3).

For Coordinates D(0, 0, 0)

Let the new coordinates of D after scaling = (X_{new} , Y_{new} , Z_{new}).

Applying the scaling equations, we have-

- $X_{\text{new}} = X_{\text{old}} \times S_x = 0 \times 2 = 0$
- $Y_{\text{new}} = Y_{\text{old}} \times S_y = 0 \times 3 = 0$
- $Z_{\text{new}} = Z_{\text{old}} \times S_z = 0 \times 3 = 0$

Thus, New coordinates of corner D after scaling = (0, 0, 0).

3D Reflection in Computer Graphics-

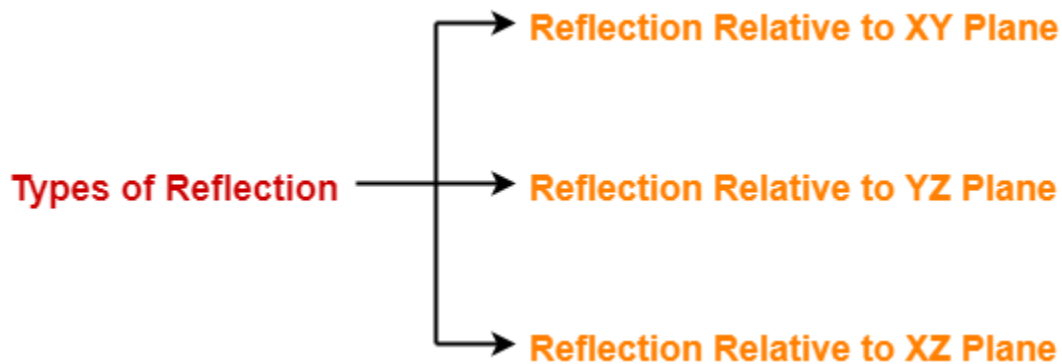
- Reflection is a kind of rotation where the angle of rotation is 180 degree.
- The reflected object is always formed on the other side of mirror.
- The size of reflected object is same as the size of original object.

Consider a point object O has to be reflected in a 3D plane.

Let-

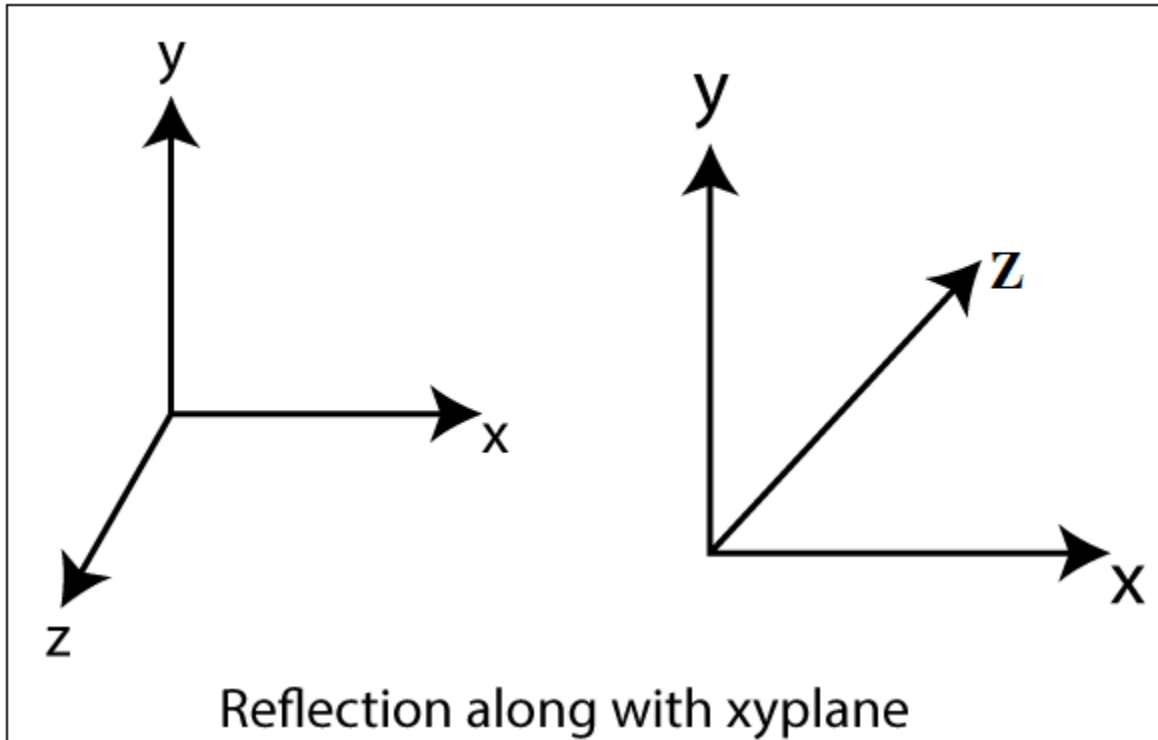
- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- New coordinates of the reflected object O after reflection = $(X_{new}, Y_{new}, Z_{new})$

In 3 dimensions, there are 3 possible types of reflection-



- Reflection relative to XY plane
- Reflection relative to YZ plane
- Reflection relative to XZ plane

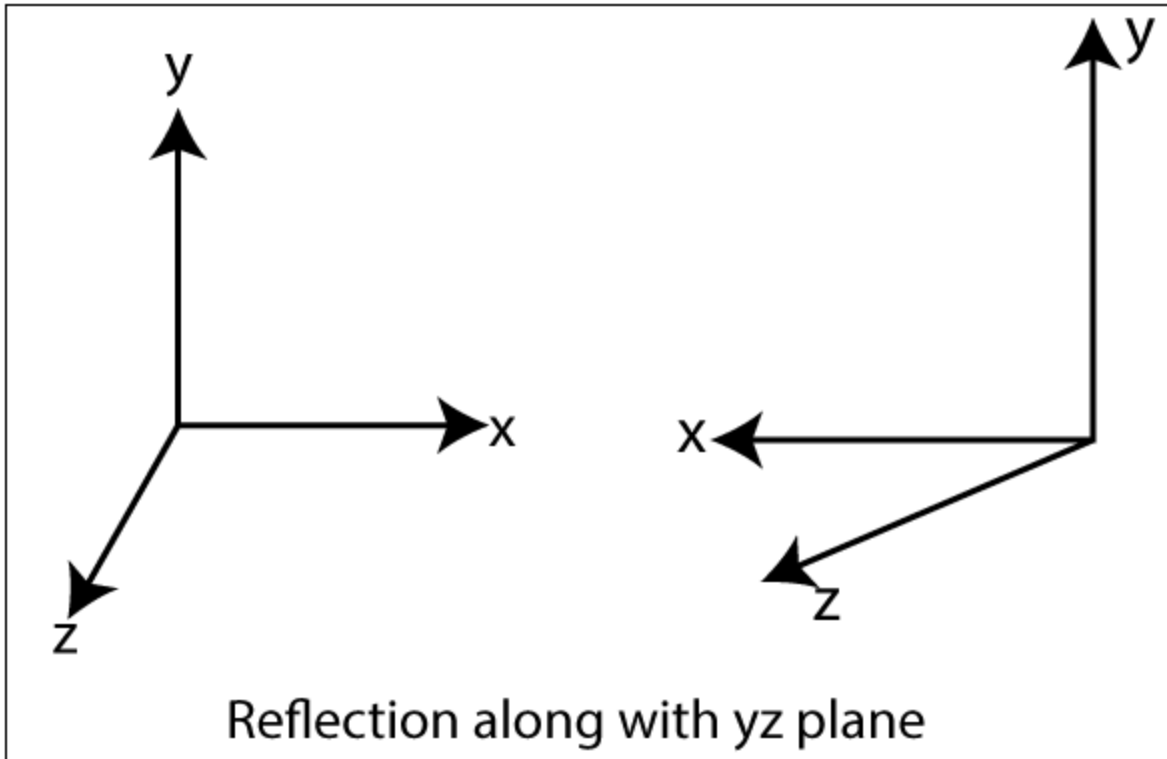
Reflection Relative to XY Plane:



This reflection is achieved by using the following reflection equations-

- $X_{\text{new}} = X_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}}$
- $Z_{\text{new}} = -Z_{\text{old}}$

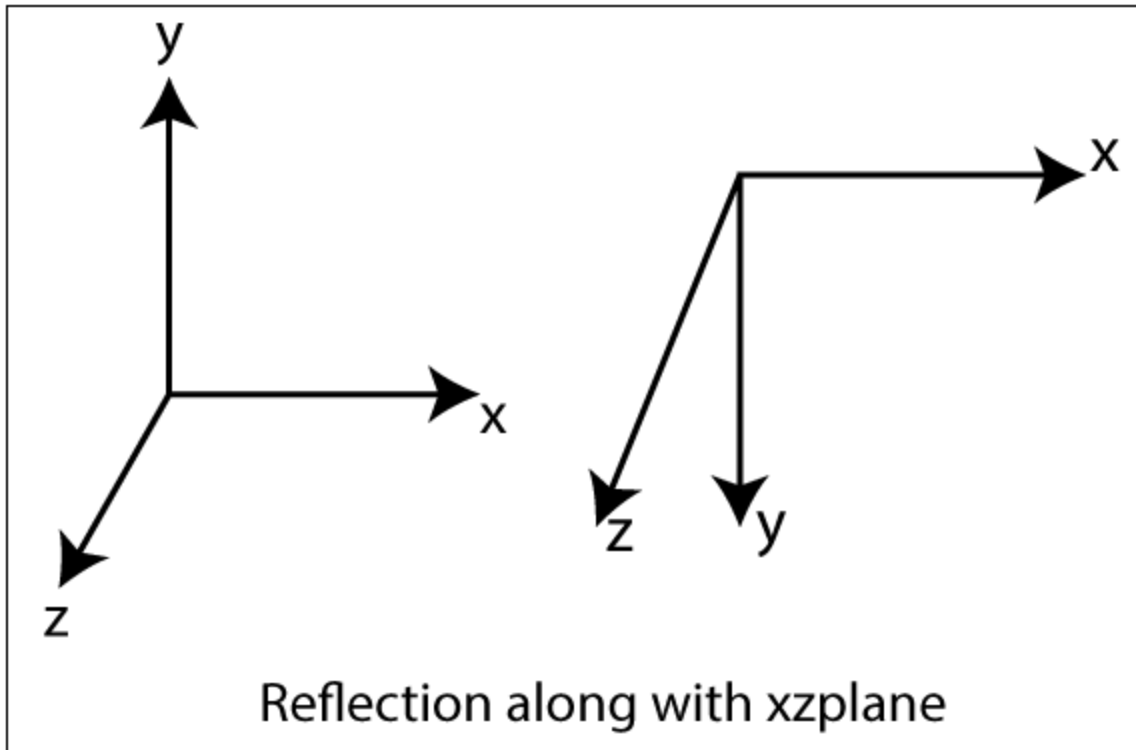
Reflection Relative to YZ Plane:



This reflection is achieved by using the following reflection equations-

- $X_{\text{new}} = -X_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}}$

Reflection Relative to XZ Plane:



This reflection is achieved by using the following reflection equations-

- $X_{\text{new}} = X_{\text{old}}$
- $Y_{\text{new}} = -Y_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}}$

Problem-01:

Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XY plane and find out the new coordinates of the object.

Solution-

Given-

- Old corner coordinates of the triangle = A (3, 4, 1), B(6, 4, 2), C(5, 6, 3)
- Reflection has to be taken on the XY plane

For Coordinates A(3, 4, 1)

Let the new coordinates of corner A after reflection = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 3$
- $Y_{\text{new}} = Y_{\text{old}} = 4$
- $Z_{\text{new}} = -Z_{\text{old}} = -1$

Thus, New coordinates of corner A after reflection = (3, 4, -1).

For Coordinates B(6, 4, 2)

Let the new coordinates of corner B after reflection = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 6$
- $Y_{\text{new}} = Y_{\text{old}} = 4$
- $Z_{\text{new}} = -Z_{\text{old}} = -2$

Thus, New coordinates of corner B after reflection = (6, 4, -2).

For Coordinates C(5, 6, 3)

Let the new coordinates of corner C after reflection = (X_{new} , Y_{new} , Z_{new}).

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 5$
- $Y_{\text{new}} = Y_{\text{old}} = 6$
- $Z_{\text{new}} = -Z_{\text{old}} = -3$

Thus, New coordinates of corner C after reflection = (5, 6, -3).

Thus, New coordinates of the triangle after reflection = A (3, 4, -1), B(6, 4, -2), C(5, 6, -3).

Problem-02:

Given a 3D triangle with coordinate points A(3, 4, 1), B(6, 4, 2), C(5, 6, 3). Apply the reflection on the XZ plane and find out the new coordinates of the object.

Solution-

Given-

- Old corner coordinates of the triangle = A (3, 4, 1), B(6, 4, 2), C(5, 6, 3)
- Reflection has to be taken on the XZ plane

For Coordinates A(3, 4, 1)

Let the new coordinates of corner A after reflection = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 3$
- $Y_{\text{new}} = -Y_{\text{old}} = -4$
- $Z_{\text{new}} = Z_{\text{old}} = 1$

Thus, New coordinates of corner A after reflection = (3, -4, 1).

For Coordinates B(6, 4, 2)

Let the new coordinates of corner B after reflection = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 6$
- $Y_{\text{new}} = -Y_{\text{old}} = -4$
- $Z_{\text{new}} = Z_{\text{old}} = 2$

Thus, New coordinates of corner B after reflection = (6, -4, 2).

For Coordinates C(5, 6, 3)

Let the new coordinates of corner C after reflection = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the reflection equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 5$
- $Y_{\text{new}} = -Y_{\text{old}} = -6$
- $Z_{\text{new}} = Z_{\text{old}} = 3$

Thus, New coordinates of corner C after reflection = (5, -6, 3).

Thus, New coordinates of the triangle after reflection = A (3, -4, 1), B(6, -4, 2), C(5, -6, 3).

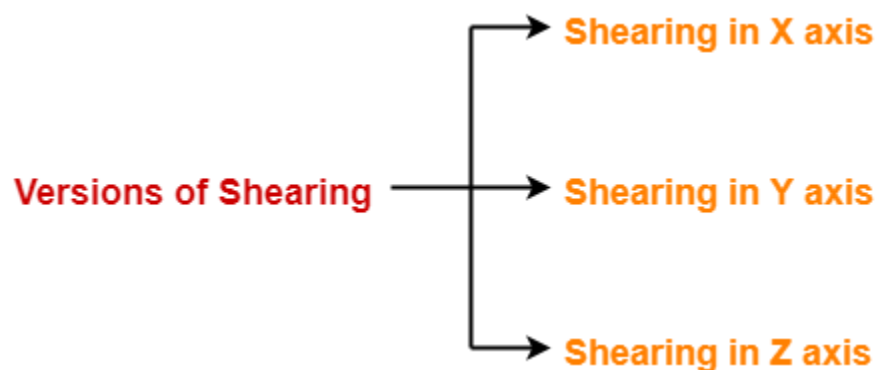
3D Shearing in Computer Graphics-

In Computer graphics,

3D Shearing is an ideal technique to change the shape of an existing object in a three dimensional plane.

In a three dimensional plane, the object size can be changed along X direction, Y direction as well as Z direction.

So, there are three versions of shearing-

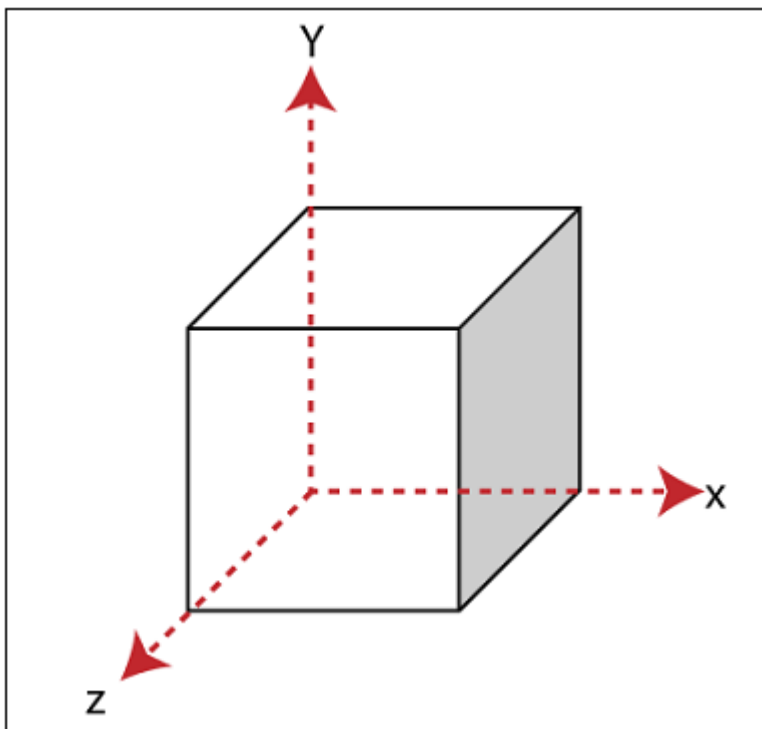


1. Shearing in X direction
2. Shearing in Y direction
3. Shearing in Z direction

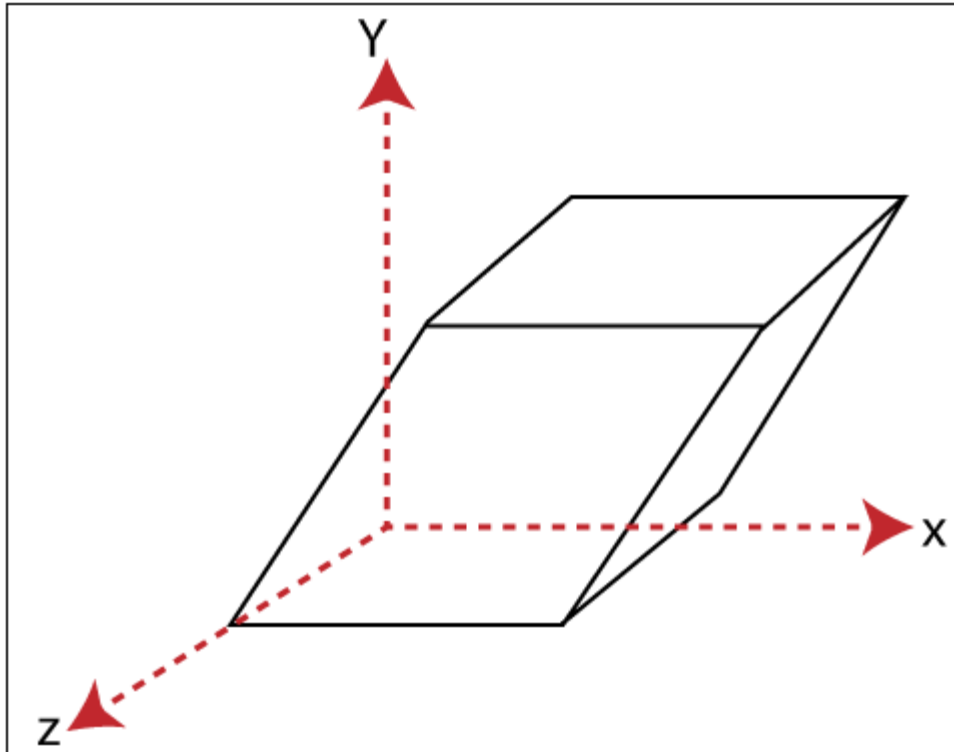
Consider a point object O has to be sheared in a 3D plane.

Let-

- Initial coordinates of the object O = $(X_{old}, Y_{old}, Z_{old})$
- Shearing parameter towards X direction = Sh_x
- Shearing parameter towards Y direction = Sh_y
- Shearing parameter towards Z direction = Sh_z
- New coordinates of the object O after shearing = $(X_{new}, Y_{new}, Z_{new})$



Shearing in X Axis-



Shearing in X axis is achieved by using the following shearing equations-

- $X_{\text{new}} = X_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times X_{\text{old}}$

Shearing in Y Axis-

Shearing in Y axis is achieved by using the following shearing equations-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Y_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times Y_{\text{old}}$

Shearing in Z Axis-

Shearing in Z axis is achieved by using the following shearing equations-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Z_{\text{old}}$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times Z_{\text{old}}$
- $Z_{\text{new}} = Z_{\text{old}}$

PRACTICE PROBLEMS BASED ON 3D SHEARING IN COMPUTER GRAPHICS-

Problem-01:

Given a 3D triangle with points (0, 0, 0), (1, 1, 2) and (1, 1, 3). Apply shear parameter 2 on X axis, 2 on Y axis and 3 on Z axis and find out the new coordinates of the object.

Solution-

Given-

- Old corner coordinates of the triangle = A (0, 0, 0), B(1, 1, 2), C(1, 1, 3)
- Shearing parameter towards X direction (Sh_x) = 2
- Shearing parameter towards Y direction (Sh_y) = 2
- Shearing parameter towards Z direction (Sh_z) = 3

Shearing in X Axis-

For Coordinates A(0, 0, 0)

Let the new coordinates of corner A after shearing = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 0$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 0 + 2 \times 0 = 0$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times X_{\text{old}} = 0 + 3 \times 0 = 0$

Thus, New coordinates of corner A after shearing = $(0, 0, 0)$.

For Coordinates B(1, 1, 2)

Let the new coordinates of corner B after shearing = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 1$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 1 + 2 \times 1 = 3$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times X_{\text{old}} = 2 + 3 \times 1 = 5$

Thus, New coordinates of corner B after shearing = $(1, 3, 5)$.

For Coordinates C(1, 1, 3)

Let the new coordinates of corner C after shearing = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} = 1$

- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times X_{\text{old}} = 1 + 2 \times 1 = 3$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times X_{\text{old}} = 3 + 3 \times 1 = 6$

Thus, New coordinates of corner C after shearing = (1, 3, 6).

Thus, New coordinates of the triangle after shearing in X axis = A (0, 0, 0), B(1, 3, 5), C(1, 3, 6).

Shearing in Y Axis-

For Coordinates A(0, 0, 0)

Let the new coordinates of corner A after shearing = (X_{new} , Y_{new} , Z_{new}).

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Y_{\text{old}} = 0 + 2 \times 0 = 0$
- $Y_{\text{new}} = Y_{\text{old}} = 0$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times Y_{\text{old}} = 0 + 3 \times 0 = 0$

Thus, New coordinates of corner A after shearing = (0, 0, 0).

For Coordinates B(1, 1, 2)

Let the new coordinates of corner B after shearing = (X_{new} , Y_{new} , Z_{new}).

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Y_{\text{old}} = 1 + 2 \times 1 = 3$
- $Y_{\text{new}} = Y_{\text{old}} = 1$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times Y_{\text{old}} = 2 + 3 \times 1 = 5$

Thus, New coordinates of corner B after shearing = (3, 1, 5).

For Coordinates C(1, 1, 3)

Let the new coordinates of corner C after shearing = (X_{new} , Y_{new} , Z_{new}).

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Y_{\text{old}} = 1 + 2 \times 1 = 3$
- $Y_{\text{new}} = Y_{\text{old}} = 1$
- $Z_{\text{new}} = Z_{\text{old}} + Sh_z \times Y_{\text{old}} = 3 + 3 \times 1 = 6$

Thus, New coordinates of corner C after shearing = (3, 1, 6).

Thus, New coordinates of the triangle after shearing in Y axis = A (0, 0, 0), B(3, 1, 5), C(3, 1, 6).

Shearing in Z Axis-

For Coordinates A(0, 0, 0)

Let the new coordinates of corner A after shearing = (X_{new} , Y_{new} , Z_{new}).

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Z_{\text{old}} = 0 + 2 \times 0 = 0$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times Z_{\text{old}} = 0 + 2 \times 0 = 0$
- $Z_{\text{new}} = Z_{\text{old}} = 0$

Thus, New coordinates of corner A after shearing = (0, 0, 0).

For Coordinates B(1, 1, 2)

Let the new coordinates of corner B after shearing = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Z_{\text{old}} = 1 + 2 \times 2 = 5$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times Z_{\text{old}} = 1 + 2 \times 2 = 5$
- $Z_{\text{new}} = Z_{\text{old}} = 2$

Thus, New coordinates of corner B after shearing = (5, 5, 2).

For Coordinates C(1, 1, 3)

Let the new coordinates of corner C after shearing = $(X_{\text{new}}, Y_{\text{new}}, Z_{\text{new}})$.

Applying the shearing equations, we have-

- $X_{\text{new}} = X_{\text{old}} + Sh_x \times Z_{\text{old}} = 1 + 2 \times 3 = 7$
- $Y_{\text{new}} = Y_{\text{old}} + Sh_y \times Z_{\text{old}} = 1 + 2 \times 3 = 7$
- $Z_{\text{new}} = Z_{\text{old}} = 3$

Thus, New coordinates of corner C after shearing = (7, 7, 3).

Thus, New coordinates of the triangle after shearing in Z axis = A (0, 0, 0), B(5, 5, 2), C(7, 7, 3).